EmotiChat - Project Source Code

1. EmotiChatServer.java

```
import com.sun.net.httpserver.HttpExchange;
import com.sun.net.httpserver.HttpServer;

import java.io.IOException;
import java.net.InetSocketAddress;
import java.nio.charset.StandardCharsets;

public class EmotiChatServer {

    public static void main(String[] args) throws IOException {

        HttpServer server = HttpServer.create(
                new InetSocketAddress(8080), 0
        );

        server.createContext("/chat", EmotiChatServer::handleChat);

        System.out.println("EmotiChat Java Server started!");
        System.out.println("Server running at http://localhost:8080");

        server.start();
    }

    private static void handleChat(HttpExchange exchange)
            throws IOException {

        exchange.getResponseHeaders().set(
                "Access-Control-Allow-Origin", "*"
        );

        exchange.getResponseHeaders().set(
                "Access-Control-Allow-Methods",
```

```
        "POST, OPTIONS"
);

exchange.getResponseHeaders().set(
        "Access-Control-Allow-Headers",
        "Content-Type"
);

if ("OPTIONS".equalsIgnoreCase(
        exchange.getRequestMethod())) {

    exchange.sendResponseHeaders(204, -1);
    exchange.close();
    return;
}

if ("POST".equalsIgnoreCase(
        exchange.getRequestMethod())) {

    String message = new String(
            exchange.getRequestBody().readAllBytes(),
            StandardCharsets.UTF_8
    );

    String emotion = detectEmotion(message);
    String response = generateResponse(emotion);

    String jsonResponse =
            "{"
            + "\"emotion\":\"" + emotion + "\","
            + "\"response\":\"" + escapeJson(response) + "\""
            + "}";

    byte[] responseBytes =
            jsonResponse.getBytes(
                    StandardCharsets.UTF_8
```

```
                );

            exchange.getResponseHeaders().set(
                    "Content-Type",
                    "application/json; charset=UTF-8"
            );

            exchange.sendResponseHeaders(
                    200,
                    responseBytes.length
            );

            exchange.getResponseBody().write(
                    responseBytes
            );

            exchange.getResponseBody().close();

        } else {

            exchange.sendResponseHeaders(405, -1);
            exchange.close();
        }
    }

    private static String detectEmotion(String message) {

        message = message.toLowerCase();

        if (message.contains("happy") ||
            message.contains("great") ||
            message.contains("excited") ||
            message.contains("good") ||
            message.contains("awesome") ||
            message.contains("wonderful") ||
            message.contains("amazing") ||
```

```
    message.contains("joy") ||
    message.contains("love") ||
    message.contains("fun") ||
    message.contains("success") ||
    message.contains("congratulations") ||
    message.contains("glad")) {

    return "HAPPY";
}

if (message.contains("sad") ||
    message.contains("cry") ||
    message.contains("lonely") ||
    message.contains("upset") ||
    message.contains("depressed") ||
    message.contains("hurt") ||
    message.contains("broken") ||
    message.contains("miss") ||
    message.contains("disappointed") ||
    message.contains("hopeless")) {

    return "SAD";
}

if (message.contains("angry") ||
    message.contains("mad") ||
    message.contains("hate") ||
    message.contains("frustrated") ||
    message.contains("annoyed") ||
    message.contains("irritated") ||
    message.contains("furious") ||
    message.contains("rage")) {

    return "ANGRY";
}
if (message.contains("scared") ||
```

```
        message.contains("afraid") ||
        message.contains("fear") ||
        message.contains("worried") ||
        message.contains("nervous") ||
        message.contains("anxious") ||
        message.contains("stress") ||
        message.contains("tension") ||
        message.contains("exam") ||
        message.contains("panic")) {

        return "FEAR";
    }

    return "NEUTRAL";
}

private static String generateResponse(String emotion) {

    switch (emotion) {

        case "HAPPY":
            return "I'm glad you're feeling happy! Keep enjoying the moment!";

        case "SAD":
            return "I'm sorry you're feeling sad. I'm here to listen.";

        case "ANGRY":
            return "I understand that you're frustrated. Take a deep breath and tell me what happened.";

        case "FEAR":
            return "It's okay to feel worried sometimes. You're not alone.";

        default:
            return "I understand. Tell me more about how you're feeling.";
    }
}
```

```
    private static String escapeJson(String text) {

        return text
                .replace("\\", "\\\\")
                .replace("\"", "\\\"")
                .replace("\n", "\\n")
                .replace("\r", "\\r");
    }
}
```

2. index.html

```
<!DOCTYPE html>
<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport"
          content="width=device-width, initial-scale=1.0">

    <title>EmotiChat</title>

    <link rel="stylesheet"
          href="/static/style.css">

</head>

<body>


<!-- ================= LOGIN PAGE ================= -->

<div id="loginPage" class="login-page">

    <div class="login-card">

        <div class="login-logo">
            [BOT]
        </div>

        <h1>EmotiChat</h1>

        <p class="login-subtitle">
            Emotion-Aware Conversational Assistant
        </p>
```

```
<h2>Welcome Back!</h2>

<p class="login-description">
    Sign in to continue your emotional conversation.
</p>


<div class="form-group">

    <label for="username">
        Username
    </label>

    <input
        type="text"
        id="username"
        placeholder="Enter your username"
        autocomplete="off"
    >

</div>


<div class="form-group">

    <label for="password">
        Password
    </label>

    <input
        type="password"
        id="password"
        placeholder="Enter your password"
    >

</div>
```

```
<div class="login-options">

    <label>

        <input
            type="checkbox"
            id="rememberMe"
        >

        Remember me

    </label>

    <span>
        EmotiChat
    </span>

</div>


<p id="loginError"
   class="login-error">
</p>


<button
    id="loginButton"
    class="login-button">

    Login

</button>


<p class="login-footer">
    Emotion Recognition - Adaptive Response
```

```
        </p>

    </div>

</div>



<!-- ================= DASHBOARD ================= -->

<div id="dashboardPage"
     class="dashboard-page"
     style="display: none;">


    <!-- SIDEBAR -->

    <aside class="sidebar">

        <div class="brand">

            <div class="brand-logo">
                [BOT]
            </div>

            <div>

                <h2>EmotiChat</h2>

                <span>
                    Emotion-Aware AI
                </span>

            </div>

        </div>
```

```
<!-- Navigation -->

<nav class="sidebar-nav">

    <button
        class="nav-item active"
        onclick="showSection('chatSection', this)">

        [CHAT]
        <span>Chat</span>

    </button>


    <button
        class="nav-item"
        onclick="showSection('emotionSection', this)">

        [BRAIN]
        <span>Emotion Detection</span>

    </button>


    <button
        class="nav-item"
        onclick="showSection('historySection', this)">

        [CHART]
        <span>Emotion History</span>

    </button>


    <button
        class="nav-item"
```

```
            onclick="showSection('profileSection', this)">

            [USER]
            <span>Profile</span>

        </button>


        <button
            class="nav-item"
            onclick="showSection('settingsSection', this)">

            [SETTINGS]?
            <span>Settings</span>

        </button>

    </nav>


    <!-- Logout -->

    <button
        class="logout-button"
        onclick="logoutUser()">

        [LOGOUT]
        <span>Logout</span>

    </button>

</aside>



<!-- MAIN CONTENT -->
```

```
<main class="main-content">


    <!-- ================= CHAT ================= -->

    <section
        id="chatSection"
        class="dashboard-section">


        <div class="top-bar">

            <div>

                <h1>
                    Chat with EmotiChat
                </h1>

                <p>
                    Share your feelings... I'm here to listen!
                </p>

            </div>


            <div class="user-badge">

                [USER]

                <span id="welcomeUser">
                    Hello, User
                </span>

            </div>

        </div>
```

```
<div class="chat-container">


    <div
        id="chatBox"
        class="chat-box">


        <div class="bot-message">

            [WAVE] Hello! I'm EmotiChat.
            How are you feeling today?

        </div>


        <div
            id="typingIndicator"
            class="bot-message typing"
            style="display: none;">

            [BOT] EmotiChat is typing...

        </div>

    </div>


    <div class="input-area">


        <div class="input-wrapper">

            <input
                type="text"
                id="messageInput"
```

```
                        placeholder="Type your message..."
                        autocomplete="off"
                    >

                    <small id="charCounter">
                        0 / 500
                    </small>

                </div>


                <button id="sendButton">

                    -> Send

                </button>

            </div>


        </div>

    </section>



    <!-- ================= EMOTION DETECTION ================= -->

    <!-- ================= EMOTION HISTORY ================= -->

<section
    id="historySection"
    class="dashboard-section"
    style="display: none;">

    <div class="page-heading">
```

```
    <h1>
        [CHART] Emotion History
    </h1>

    <p>
        Overview of your previous emotional interactions.
    </p>

</div>


<!-- Statistics -->

<div class="stats-grid">

    <div class="stat-card">

        <span class="stat-icon">
            [CHAT]
        </span>

        <div>

            <p>
                Total Messages
            </p>

            <h2 id="totalMessages">
                0
            </h2>

        </div>

    </div>
    <div class="stat-card">
        <span
```

```
            class="stat-icon"
            id="frequentEmotionIcon">

            [HAPPY]

        </span>

        <div>

            <p>
                Most Frequent Emotion
            </p>

            <h2 id="frequentEmotion">
                ?
            </h2>

        </div>

    </div>

</div>


<!-- History Content -->

<div class="history-grid">


    <!-- Emotion Distribution -->

    <div class="history-card">

        <h2>
            Emotion Distribution
        </h2>
```

```
<div class="emotion-bar-item">

    <div class="emotion-bar-label">

        <span>
            [HAPPY] Happy
        </span>

        <strong id="happyCount">
            0
        </strong>

    </div>

    <div class="progress-background">

        <div
            id="happyBar"
            class="progress-bar happy-bar">
        </div>

    </div>

</div>


<div class="emotion-bar-item">

    <div class="emotion-bar-label">

        <span>
            [SAD] Sad
        </span>

        <strong id="sadCount">
            0
```

```
        </strong>

    </div>

    <div class="progress-background">

        <div
            id="sadBar"
            class="progress-bar sad-bar">
        </div>

    </div>

</div>


<div class="emotion-bar-item">

    <div class="emotion-bar-label">

        <span>
            [ANGRY] Angry
        </span>

        <strong id="angryCount">
            0
        </strong>

    </div>

    <div class="progress-background">

        <div
            id="angryBar"
            class="progress-bar angry-bar">
        </div>
```

```
    </div>

</div>


<div class="emotion-bar-item">

    <div class="emotion-bar-label">

        <span>
            [FEAR] Fear
        </span>

        <strong id="fearCount">
            0
        </strong>

    </div>

    <div class="progress-background">

        <div
            id="fearBar"
            class="progress-bar fear-bar">
        </div>

    </div>

</div>


<div class="emotion-bar-item">

    <div class="emotion-bar-label">

        <span>
```

```
                [NEUTRAL] Neutral
            </span>

            <strong id="neutralCount">
                0
            </strong>

        </div>

        <div class="progress-background">

            <div
                id="neutralBar"
                class="progress-bar neutral-bar">
            </div>

        </div>

    </div>

</div>


<!-- Recent Emotions -->

<div class="history-card">

    <h2>
        Recent Emotions
    </h2>


    <div
        id="recentEmotions"
        class="recent-emotions">
        <div class="empty-history">
```

```
                    No emotional interactions yet.

                </div>

            </div>

        </div>

    </div>


    <!-- Clear History -->

    <button
        class="clear-history-button"
        onclick="clearEmotionHistory()">

        [DELETE]? Clear Emotion History

    </button>

</section>

        <!-- ================= HISTORY ================= -->

        <section
            id="historySection"
            class="dashboard-section"
            style="display: none;">


            <div class="page-heading">

                <h1>
                    [CHART] Emotion History
                </h1>
```

```
    <p>
        Overview of your previous emotional interactions.
    </p>

</div>


<div class="stats-grid">


    <div class="stat-card">

        <span class="stat-icon">
            [CHAT]
        </span>

        <div>

            <p>
                Total Messages
            </p>

            <h2 id="totalMessages">
                0
            </h2>

        </div>

    </div>


    <div class="stat-card">

        <span class="stat-icon">
            [HAPPY]
        </span>
```

```
        <div>

            <p>
                Most Frequent Emotion
            </p>

            <h2 id="frequentEmotion">
                ?
            </h2>

        </div>

    </div>


</div>


<div class="history-card">

    <h2>
        Emotion Distribution
    </h2>

    <div class="emotion-list">

        <div>
            <span>[HAPPY] Happy</span>
            <strong id="happyCount">0</strong>
        </div>

        <div>
            <span>[SAD] Sad</span>
            <strong id="sadCount">0</strong>
        </div>
        <div>
```

```
                        <span>[ANGRY] Angry</span>
                        <strong id="angryCount">0</strong>
                    </div>

                    <div>
                        <span>[FEAR] Fear</span>
                        <strong id="fearCount">0</strong>
                    </div>

                    <div>
                        <span>[NEUTRAL] Neutral</span>
                        <strong id="neutralCount">0</strong>
                    </div>

                    <div class="analytics-card">

    <h3>[CHART] Emotion Analytics</h3>

    <p>
        Visual representation of your emotional interactions.
    </p>

    <div class="chart-container">
        <canvas id="emotionChart"></canvas>
    </div>

</div>
                </div>

            </div>

        </section>


        <!-- ================= PROFILE ================= -->
<section
```

```
id="profileSection"
class="dashboard-section"
style="display: none;">

<div class="page-heading">

    <h1>[USER] Profile</h1>

    <p>
        Manage your EmotiChat profile information.
    </p>

</div>


<div class="profile-card">

    <div class="profile-avatar">
        [USER]
    </div>

    <h2 id="profileUsername">
        User
    </h2>

    <p class="profile-role">
        EmotiChat User
    </p>


    <div class="profile-info">

        <div>

            <span>Username</span>
            <strong id="profileName">
```

```
        User
    </strong>

</div>


<div>

    <span>Application</span>

    <strong>
        EmotiChat
    </strong>

</div>


<div>

    <span>Account Type</span>

    <strong>
        Standard User
    </strong>

</div>


<div>

    <span>Detection System</span>

    <strong>
        Keyword-Based
    </strong>
</div>
```

```
        </div>

    </div>

</section>


        <!-- ================= SETTINGS ================= -->

<section
    id="settingsSection"
    class="dashboard-section"
    style="display: none;">

    <div class="page-heading">

        <h1>[SETTINGS]? Settings</h1>

        <p>
            Customize your EmotiChat experience.
        </p>

    </div>


    <div class="settings-card">


        <!-- Enter Key -->

        <div class="setting-row">

            <div>

                <strong>
                    Enter key to send
```

```
        </strong>

        <small>
            Press Enter to send your message.
        </small>

    </div>

    <input
        type="checkbox"
        id="enterKeySetting"
        checked
    >

</div>


<!-- Typing Indicator -->

<div class="setting-row">

    <div>

        <strong>
            Typing indicator
        </strong>

        <small>
            Show EmotiChat typing animation.
        </small>

    </div>

    <input
        type="checkbox"
        id="typingSetting"
```

```
        checked
    >

</div>


<!-- Character Counter -->

<div class="setting-row">

    <div>

        <strong>
            Character counter
        </strong>

        <small>
            Display the 500-character limit.
        </small>

    </div>

    <input
        type="checkbox"
        id="counterSetting"
        checked
    >

</div>


<!-- Clear History -->

<div class="setting-row">

    <div>
```

```
                <strong>
                    Clear emotion history
                </strong>

                <small>
                    Remove all saved emotion records.
                </small>

            </div>

            <button
                class="settings-clear-button"
                onclick="clearEmotionHistory()">

                Clear

            </button>

        </div>


    </div>


    <p
        id="settingsMessage"
        class="settings-message">
    </p>

</section>


<!-- ================ DASHBOARD JAVASCRIPT ================ -->

<script>
    const loginPage =
```

```
    document.getElementById("loginPage");

const dashboardPage =
    document.getElementById("dashboardPage");

const loginButton =
    document.getElementById("loginButton");

const username =
    document.getElementById("username");

const password =
    document.getElementById("password");

const loginError =
    document.getElementById("loginError");


/*
 * Check previous login
 */

if (
    sessionStorage.getItem("emotichatLoggedIn")
    === "true"
) {

    showDashboard();

}


/*
 * Login
 */
loginButton.addEventListener(
```

```
"click",
function () {

    const user =
        username.value.trim();

    const pass =
        password.value.trim();


    if (
        user === "" ||
        pass === ""
    ) {

        loginError.textContent =
            "[WARNING]? Please enter username and password.";

        return;

    }


    /*
     * Demo credentials
     *
     * Username: admin
     * Password: 1234
     */

    if (
        user === "admin" &&
        pass === "1234"
    ) {

        sessionStorage.setItem(
```

```
                "emotichatLoggedIn",
                "true"
            );

            sessionStorage.setItem(
                "emotichatUsername",
                user
            );

            loginError.textContent = "";

            showDashboard();

        } else {

            loginError.textContent =
                "[ERROR] Invalid username or password.";

        }

    }
);


/*
 * Enter key on password
 */

password.addEventListener(
    "keypress",
    function (event) {

        if (event.key === "Enter") {

            loginButton.click();
        }
```

```
    }
);


/*
 * Show Dashboard
 */

function showDashboard() {

    loginPage.style.display =
        "none";

    dashboardPage.style.display =
        "flex";


    const savedUser =
        sessionStorage.getItem(
            "emotichatUsername"
        ) || "User";


    document.getElementById(
        "welcomeUser"
    ).textContent =
        "Hello, " + savedUser;


    document.getElementById(
        "profileUsername"
    ).textContent =
        savedUser;

    document.getElementById(
        "profileName"
```

```
    ).textContent =
        savedUser;

}


/*
 * Navigation
 */

function showSection(
    sectionId,
    clickedButton
) {

    const sections =
        document.querySelectorAll(
            ".dashboard-section"
        );


    sections.forEach(
        function (section) {

            section.style.display =
                "none";

        }
    );


    document.getElementById(
        sectionId
    ).style.display =
        "block";
    const navItems =
```

```
        document.querySelectorAll(
            ".nav-item"
        );


    navItems.forEach(
        function (item) {

            item.classList.remove(
                "active"
            );

        }
    );


    clickedButton.classList.add(
        "active"
    );

}


/*
 * Logout
 */

function logoutUser() {

    sessionStorage.removeItem(
        "emotichatLoggedIn"
    );

    sessionStorage.removeItem(
        "emotichatUsername"
    );
```

```
        dashboardPage.style.display =
            "none";

        loginPage.style.display =
            "flex";


        username.value = "";

        password.value = "";

        loginError.textContent = "";

    }

</script>


<!-- Existing Chat JavaScript -->
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script src="/static/script.js"></script>


</body>

</html>
```